Electrical Systems

Kirchhoff's laws to transfer functions, and the electrical-mechanical analogy.

Ogata, *Modern Control Engineering*, Ch. 3.

In this tutorial you will:

- model a series RLC circuit with KVL,
- see the force-voltage analogy with the mass-spring-damper, and
- model an op-amp PI controller circuit.

Step through with **Ctrl+Enter**, or render a report with `publish`.

Electrical circuits are modeled with Kirchhoff's voltage and current laws. Consider a series RLC circuit driven by source voltage $e_i(t)$, with output taken as the voltage across the capacitor $e_o(t)$. KVL around the loop, with loop current $i(t)$:

$L\frac{di}{dt} + Ri + \frac{1}{C}\int i\,dt = e_i$

and since $e_o = \frac{1}{C}\int i\,dt$, we have $i = C\dot{e}_o$. Substituting:

$LC\ddot{e}_o + RC\dot{e}_o + e_o = e_i$

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Transfer Function of the RLC Circuit

Taking the Laplace transform with zero initial conditions:

$(LCs^2 + RCs + 1)E_o(s) = E_i(s) \quad \Rightarrow \quad G(s) = \frac{E_o(s)}{E_i(s)} = \frac{1}{LCs^2+RCs+1}$

With $L = 1\,\mathrm{H}$, $R = 2\,\Omega$, $C = 0.5\,\mathrm{F}$:

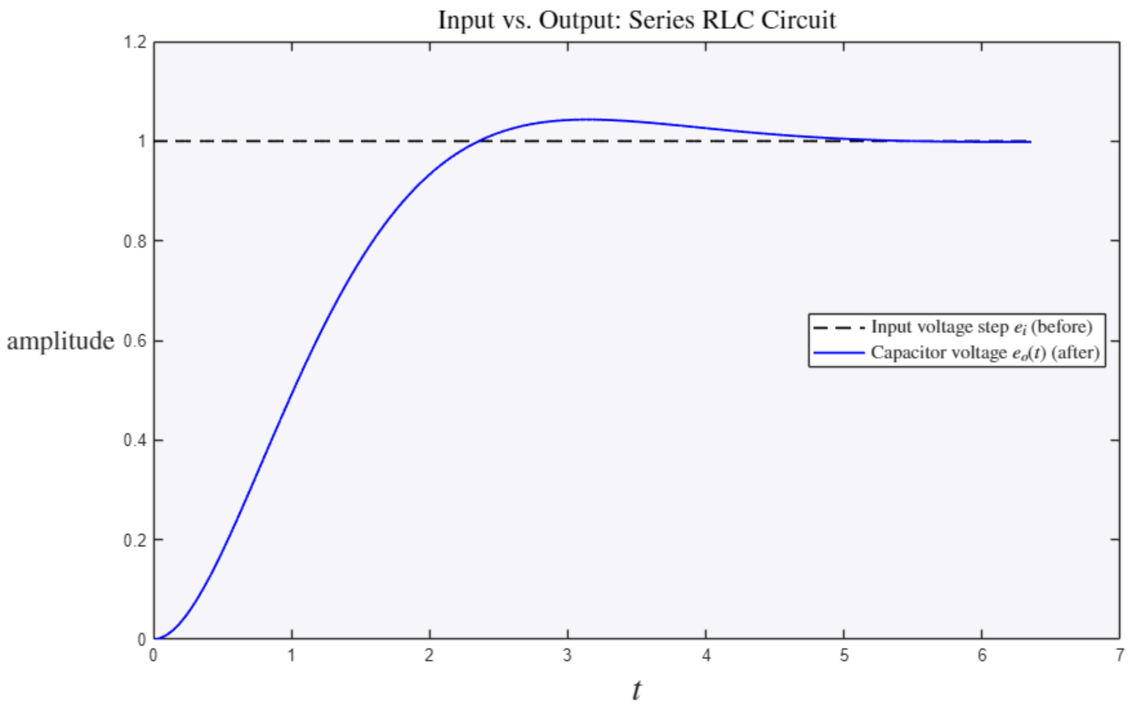
```
L = 1; R = 2; C = 0.5;
G_rlc = tf(1,[L*C R*C 1])

figure
[es,ts] = step(G_rlc);
plot(ts, ones(size(ts)),'k--', ts, es,'b','LineWidth',1.3)
legend('Input voltage step $e_i$ (before)','Capacitor voltage $e_o(t)$ (after)', ...
    'Interpreter','latex','FontSize',11,'Location','east')
title('Input vs. Output: Series RLC Circuit','Interpreter','latex','FontSize',16)
ylabel('amplitude','Interpreter','latex','FontSize',16)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$','Interpreter','latex','FontSize',20)
```

```
G_rlc =

         1
  ---------------
  0.5 s^2 + s + 1

Continuous-time transfer function.
```



The Electrical-Mechanical Analogy

The RLC circuit equation is structurally identical to the mass-spring-damper equation $m\ddot{x} + b\dot{x} + kx = u$ from `MechanicalSystems.m`, under the force-voltage analogy:

```
Mechanical          Electrical
------------------  ------------------
Force, u            Voltage, e
Mass, m             Inductance, L
Damping coeff., b   Resistance, R
Spring const., k    Elastance, 1/C
Displacement, x      Charge, q
Velocity, dx/dt      Current, i
```

This analogy is why the same transfer-function machinery (and the same MATLAB code) applies to both physical domains.

Operational-Amplifier Circuit

A common building block is the inverting op-amp circuit with input impedance $Z_i(s)$ and feedback impedance $Z_f(s)$, giving

$\frac{E_o(s)}{E_i(s)} = -\frac{Z_f(s)}{Z_i(s)}$

For a PI-controller circuit, $Z_i = R_1$ (a resistor) and $Z_f = R_2 + \frac{1}{C_2 s}$ (a resistor and capacitor in series):

$G(s) = -\frac{R_2+\frac{1}{C_2 s}}{R_1} = -\frac{R_2 C_2 s+1}{R_1 C_2 s}$

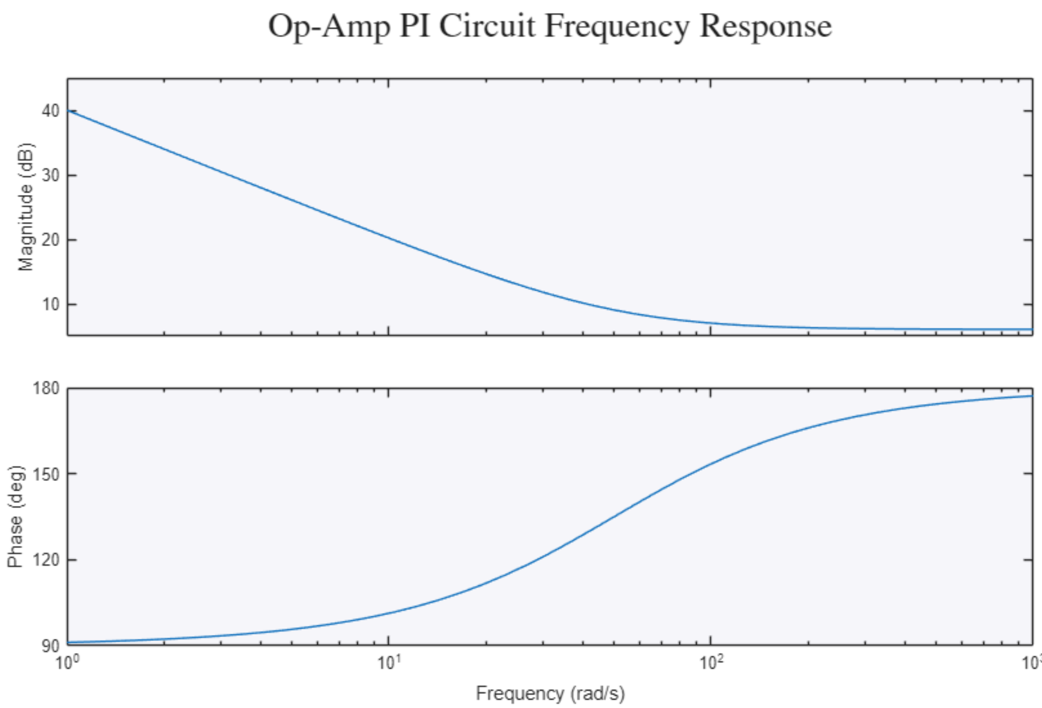
```
R1 = 10e3; R2 = 20e3; C2 = 1e-6;
G_opamp = tf(-[R2*C2 1],[R1*C2 0])

figure
bode(G_opamp)
title('Op-Amp PI Circuit Frequency Response','Interpreter','latex','FontSize',20)
```

```
G_opamp =

  -0.02 s - 1
  -----------
    0.01 s

Continuous-time transfer function.
```



Try it yourself

- Shrink `R` toward 0 and notice the RLC response become more oscillatory (less damping) – the electrical twin of a light mechanical damper.
- Check the analogy: build the mass-spring-damper with `m=L, b=R, k=1/C` and confirm its poles match the circuit.